MAYUR GANGIREDDY

AI & Machine Learning Engineer • Full-Stack & Mobile Systems

Hyderabad, Telangana, India | +91 6281192139 | mayurgangereddy12345@gmail.com | GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Machine Learning & Full-Stack Engineer with proven production experience architecting end-to-end AI systems, mobile applications deployed to the Google Play Store and Apple App Store, and scalable microservices. Specialized in time-series energy disaggregation (NILM), privacy-preserving Agentic AI via the Model Context Protocol (MCP), native hardware integrations (USB IRDA / OCR), and modern web platforms.

WORK EXPERIENCE

Bharat Smart Services – **Machine Learning & Full-Stack Engineer** 01/2026 – Present | Hyderabad, India

- **NILM Energy Disaggregation AI:** Built 5-second smart meter time-series models with **XGBoost** (benchmarked against CNNs/Transformers), achieving **97% accuracy on Geysers** and **~85% on AC, Fridge, and Washing Machines**.
- **NerveNet – Enterprise MCP Platform:** Architected zero-knowledge AI platform on **Model Context Protocol (MCP)** with client-side encryption over CSV/SQL datasets to protect private company data from cloud LLMs.
- **Bharat Smart Services App (Utility & Billing)** Google Play App Store: Built React/Capacitor app with native Java USB IRDA serial drivers, camera OCR verification review, and utility billing (TSSPDCL, TSNPDCL, HMWSSB).
- **Bharat Smart Home (IoT Automation)** Google Play App Store: Built smart home app featuring device hierarchy, condition-based automation schedules, lifestyle scenes, and real-time disaggregated energy intelligence.

Telangana State Police Academy (TGPA) – **Android Application Developer** 03/2024 – 08/2024 | Hyderabad, India

- **Criminal Laws by TGPA** Google Play Store: Designed, built, and published the official government legal repository Android application for police officers and legal trainees. Implemented sub-second offline search across criminal penal codes, bilingual document viewers, and high-performance SQLite caching.

Acmegrade – **Java Software Engineering Intern** 11/2023 – 01/2024 | Virtual

- Developed backend core Java service modules and relational database schemas with high-throughput transaction handling and structured error-handling middleware.

SELECTED PROJECTS

- **Hostel Hub Pro & Companion Mobile App:** Multi-tenant campus accommodation platform connecting administration, wardens, students, and guards. Built administrative web portal with **TanStack Start SSR & React**, companion **Flutter app** with 24-hr time-restricted cryptographic QR gate passes, and an Express/MySQL backend on Plesk.
- **Satellite Ship Detection & Classification (YOLOv8)** Live Demo GitHub: Engineered post-CNN framework for high-resolution satellite imagery analysis. Implemented automated water/land masking followed by YOLOv8 deep learning vessel detection with confidence bounding boxes.
- **Samarth – Intelligent Agriculture & Climate Q&A Engine** GitHub: Built deterministic Natural Language to SQL query translation engine over Indian agricultural datasets using FastAPI, schema-aware LLM prompt architecture, and DuckDB analytical database with 100% SQL traceability.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript (ES6+), Java, SQL, Dart, C, C++
AI & Machine Learning:	XGBoost, PyTorch, Transformers, CNNs, Time-Series NILM, Model Context Protocol (MCP), YOLOv8, Computer Vision (OCR), LangChain, Scikit-Learn
Web & Mobile:	React.js, Node.js, Express, FastAPI, Capacitor.js, Flutter, Android (Java/Kotlin), Tailwind CSS, Vite
Databases & Cloud:	PostgreSQL, MySQL, MariaDB, DuckDB, Supabase, Firebase, Plesk Linux, REST APIs
Tools & DevOps:	Git, Docker, Android Studio, Xcode, Postman, Linux CLI, USB Serial Hardware (IRDA)

EDUCATION & HONORS

Sphoorthy Engineering College – B.Tech in Computer Science & Engineering (AIML) 2021 – 2025 | Hyderabad, India

Sri Gayatri Junior College – Board of Intermediate Education (MPC) 2019 – 2021 | Hyderabad, India

National Finalist – Smart India Hackathon (SIH) **Production Apps** – 3 Published Apps on Google Play Store & Apple App Store